COMP1230 — Group Project

Study Group Finder

Weight	**25%** of the final grade, marked out of 100
Work	Teams of **3** from the same section. Every member runs **their own copy** of the website
Phases	Phase 1 (20), Phase 2 (30), Phase 3 (30), and your own video (20)
Due dates	In the table in §2. If a date here is ever different from D2L, use the date shown in D2L
Outcomes assessed	1–9
Generative AI	**Permitted**, with the AI Usage Declaration form for every phase (§9)

Your team builds a website where students find people to study with. A student registers, logs in, and creates a **study group** for a course: what it is about, when and where it meets, and how many people it can hold. Other students browse the groups and join the ones they want.

You build it in three phases. Each phase uses what the lectures and labs have just taught. The project also uses the SQL you learned in your earlier database course: the home page query uses a `JOIN`.

Some words used in this brief:

- **study group** — one row in the `study_groups` table: a course, a title, a time, a place and a capacity.
- **owner** — the user who created a study group.
- **member** — a user who has joined a study group (Phase 2).
- **your copy** — the website running on **your own** GBLearn account, with your own database.
- **teammate** — one of the other two students in your project team. (A *study group* is what the website manages. Your *team* is the three of you.)
- **the checker** — the instructor's program that tests your copy by opening its pages and sending its forms.

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1 · How the project works

- **One set of files, three copies.** Your team shares one set of PHP files. Then **each member** uploads the files to their own GBLearn account and runs them with their own database. Every copy is marked separately, so a member whose copy does not work loses those marks, and the others do not.
- **Every member submits every phase in D2L** (§8).
- **Your team's private discussion topic** in D2L is where you plan the work, share code and keep a record of who did what.
- **Divide the work so each person is responsible for specific features.** Every member must be able to explain the code they wrote. Your video (§7) is where you show it.
- **Later phases change some earlier rules.** Where a phase says so, its rule replaces an earlier one: Phase 2 makes edit and delete **owner-only**, and Phase 3 may change where files and settings are kept, while the address of your copy stays the same. Each phase is checked against its own rules only.

2 · Dates

Item	Due (11:59 p.m.)	Week	Marks
Join a team in D2L **Groups**	Sun 18 Oct	end of Week 6	—
Phase 1 · Set-up, accounts and groups	Sun 8 Nov	end of Week 9	20
Team evaluation, check-in	Sun 22 Nov	end of Week 11	—
Phase 2 · Ownership, security and joining	Sun 22 Nov	end of Week 11	30
Progress check in the lab	in the lab	Week 13	—
Phase 3 · Structure, data out, finish and deploy	Sun 13 Dec	end of Week 14	30
Video (individual)	Sun 13 Dec	end of Week 14	20
Team evaluation, final	Sun 13 Dec	end of Week 14	—

Nothing is due in the mid-term week (Week 7) or intersession week (Week 8).

Late work loses **20% of that phase's mark per day**, for up to **5 days**. Part of a day counts as a day. After 5 days the D2L folder closes.

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3 · Where your copy lives

Your project goes in the `assignments` folder you made in Lab 1:

```
public_html/comp1230/assignments/project/
```

Its address is:

```
https://<username>.gblearn.com/comp1230/assignments/project/
```

Use exactly this folder name, in lowercase. The checker and the marker start at this address. In Phases 1 and 2, all your files go in this folder. Phase 3 may move some files outside it, as taught in Weeks 11 and 14; its instructions will say exactly where.

Your database. Make a new database for the project, the same way you did before Lab 6: cPanel → **Databases** → **Manage My Databases** → create `project` (it becomes something like `f6123456_project`), add a database user to it with **SELECT, INSERT, UPDATE, DELETE**, then open phpMyAdmin, click the new database, and **Import** the supplied `schema.sql`.

`schema.sql` creates three tables and puts in some starting data:

Table	What it holds
`users`	`id`, `username` (unique), `display_name`, `password_hash`, `created_at`
`study_groups`	`id`, `owner_id` (→ `users.id`), `course_code`, `title`, `description`, `meets_at`, `location`, `capacity`, `created_at`

Table	What it holds
`memberships`	`user_id`, `group_id`, `joined_at` — one row per member of a study group. **Used from Phase 2**

Do not change the tables. The checker relies on these exact names. Importing `schema.sql` again resets your data to the start.

The test accounts and sample groups

`schema.sql` creates two accounts that the checker and the marker use to log in to your copy:

Username	Password	Display name
`tester1`	`Study-1230`	Tester One
`tester2`	`Study-1230`	Tester Two

It also creates two **sample groups**: "PDO practice session" (owner Tester One) and "HTML and CSS review" (owner Tester Two).

- **Your login page must show both accounts and the password**, in a box above or beside the form, so anyone testing your copy can log in.
- **Do not delete or change the two accounts or the two sample groups.** The checker looks for them. To test your own edit and delete pages, use groups you create yourself.
- When the checker tests your copy, it **adds its own users and study groups**. Their names start with `chk`, followed by a different code each time. Leave them; they do no harm. Do not use names starting with `chk` yourself.

4 · Phase 1 · Set-up, accounts and groups

Due Sun 8 Nov, 11:59 p.m. (end of Week 9) · 20 marks

You can start most of Phase 1 now. The database pages use Week 6 (PDO). Register, log in and log out use Week 9, which is taught on Tue 3 Nov. Leave those for last.

Everything in §4 is a **Phase 1** rule. §1 explains how later phases change some of them.

4.1 Files and pages

Use **these file names**. The checker opens them by name.

File	What it does
`db.php`	Returns the PDO connection, as in Lab 6. In Phase 1, **the only file that contains your database name, user and password**
`index.php`	The home page: the list of study groups (§4.2)
`register.php`	Create an account (§4.3)
`login.php`	Log in (§4.4). Shows the two test accounts
`logout.php`	Logs out. It accepts **POST** only
`group-create.php`	Add a study group. Logged-in users only
`group-edit.php?id=N`	Change study group N. Logged-in users only
`group-delete.php`	Delete a study group. Accepts **POST** only, with the group's `id` in a field named `id`. Logged-in users only
`team.php`	The Team page (§4.6)
`aud-phase1.pdf`	Your own AI Usage Declaration for Phase 1 (§9)

You may add any other files you need, for example a shared header and footer.

On every page:

- links to **Home** and **Team**;
- when logged out: links to **Log in** and **Register**;
- when logged in: the text **`Logged in as <display name>`** and a **Log out** button (a small POST form to `logout.php`).

Error messages go inside an element whose `class` includes `error`, next to the form. For example, `<p class="error">Capacity must be between 2 and 10.</p>` or `<div class="error message">…</div>`. The checker looks for this class.

Lengths in the rules below are counted in **characters**, with `mb_strlen()` (Week 3).

4.2 The home page

`index.php` lists **every** study group, earliest `meets_at` first, in a table with these columns:

Course · Title · When · Where · Capacity · Created by

Created by is the owner's `display_name`, so the query joins `study_groups` and `users`. As a reminder of the syntax, here is a join between two other tables:

```
SELECT books.title, authors.name
FROM books
JOIN authors ON authors.id = books.author_id
ORDER BY books.title;
```

When a user is logged in, each row also has an **Edit** link to exactly `group-edit.php?id=N` (N is that group's `id`) and a **Delete** button (a small POST form to `group-delete.php`).

After importing `schema.sql`, your home page shows the two sample groups.

4.3 Register

The form posts to `register.php`. Its fields are named `username`, `display_name`, `password` and `password_confirm`.

Field	Rule
`username`	required; 3–20 characters; only lowercase letters, digits and _; not already taken
`display_name`	required; up to 50 characters
`password`	required; at least 8 characters
`password_confirm`	must be the same as `password`

- When a value is wrong, show the form again with an error, and keep what the user typed (not the passwords). Save nothing.
- Store the password with `password_hash()`. **Never store the password itself.**
- A taken username shows the error `That username is already taken.` Do not let the database error reach the page.
- When registration succeeds, the user is logged in and sent to the home page.

4.4 Log in and log out

- The form posts to `login.php`. Its fields are named `username` and `password`.
- Check the password with `password_verify()`.
- A wrong username **and** a wrong password both show the same message: `Username or password is incorrect.`
- On success, call `session_regenerate_id(true)`, remember the user in the session, and send them to the home page.
- `logout.php` clears the session and sends the user to the home page.

4.5 Study groups

`group-create.php` and `group-edit.php` each have a form with these field names:

Field	Rule
`course_code`	required; 4 letters and 4 digits, like `COMP1230`; store it in capitals
`title`	required; 3–80 characters
`description`	required; up to 1000 characters
`meets_at`	required; a real date and time (see below). Use `<input type="datetime-local">`
`location`	required; up to 80 characters, for example `Library, room 2` or `Online`
`capacity`	required; a whole number from 2 to 10

- The create form posts to `group-create.php`. The edit form posts to `group-edit.php?id=N`, the same address that showed it.
- **Validate every field in PHP before saving.** If any field is wrong, show the same form again with an error and keep **all** the values the user typed. **Do not add a row, and do not change the existing row.**
- A logged-out visitor who opens `group-create.php`, `group-edit.php` or `group-delete.php` is **redirected to `login.php`**.
- New groups are saved with the logged-in user as `owner_id`.
- After a successful save or delete, redirect to the home page (the PRG pattern from Week 3).
- `group-edit.php?id=N` for a group that does not exist shows a "not found" message, with status 404.
- **In Phase 1, any logged-in user may edit or delete any group.** Phase 2 limits this to the owner.

The date and time. A `datetime-local` input sends text like `2026-11-20T15:00`: the date, a `T`, then hours and minutes. Check it the way Chapter 3 checks a date: the length is 16, the parts are digits, the date passes `checkdate()`, the hour is 00–23 and the minute is 00–59. Past dates are allowed. Then convert it for MySQL, and back again to fill in the edit form:

```php
$forDatabase = str_replace('T', ' ', $meetsAt) . ':00';               // '2026-11-20 15:00:00'
$forTheForm  = str_replace(' ', 'T', substr($row['meets_at'], 0, 16));  // '2026-11-20T15:00'
```

4.6 The Team page

`team.php` shows your team number and section (CRN), and one row for each of the three members:

Column	What goes in it
Member	full name, as it appears in D2L
Student ID	the member's George Brown student number
Copy	a link to that member's copy: `https://<their username>.gblearn.com/comp1230/assignments/project/`
AI declarations	a link to each of that member's `aud-phaseN.pdf` files submitted so far
Video	a link to that member's video (added in Phase 3)

All three copies have the same Team page, so opening any one copy reaches every member's copy and every declaration. Check that every link opens. The instructor checks the names and student IDs against the D2L class list. The checker does **not** open your teammates' copies, so a teammate's copy never affects your mark.

4.7 How the checker uses your copy

- It sends forms with **POST** to these addresses: `register.php`, `login.php`, `logout.php`, `group-create.php`, `group-edit.php?id=N` and `group-delete.php`.
- It sends **only** the field names in this brief. Your forms may contain other fields, but in Phase 1 your pages must work without them.
- It keeps cookies like a browser and follows redirects. Logged-out tests and the wrong-password test start with no cookies.
- It finds a group it created by its title on the home page, and reads the group's `id` from that row's **Edit** link. It edits and deletes **only** its own `chk` groups.
- It checks only what this brief describes. Layout, colours and extra features are not marked.

4.8 How Phase 1 is marked

Checklist, 16 marks. The checker runs these checks **on your copy**:

#	Check	Marks
1	`index.php` opens (status 200) and shows no PHP error or warning text	1
2	The home page lists the two sample groups (1), each with its owner's display name (1)	2
3	`team.php` shows the team number, the section and three members, each with a full name and a student ID (1); it has three Copy links in the address format of §4.6 (1)	2
4	`login.php` shows the two test accounts (1); logging in as `tester1` shows `Logged in as Tester One` (1)	2
5	A wrong password shows `Username or password is incorrect.` and does not log in	1
6	Registering a new user logs them in (1); registering the same username again shows an error (1)	2
7	A group created while logged in appears on the home page	2
8	After logging out, `group-create.php` redirects to `login.php`	1
9	Editing a group changes it on the home page	1
10	Deleting a group (POST) removes it from the home page	1

#	Check	Marks
11	An empty title (½) and a capacity of 11 (½) are each refused: an error message, the typed values kept, and no group added	1

Code, 4 marks. The marker reads your code:

Criterion	Marks
Every query uses a prepared statement, with placeholders for every value. No variable is put inside an SQL string	1
Passwords use `password_hash()` and `password_verify()`; `session_regenerate_id(true)` runs on login	1
Every value that came from the user or the database is printed with `htmlspecialchars()`	1
The database details appear only in `db.php`; the header and footer are shared, not copied into every page	1

AI declaration: a missing or unopenable `aud-phase1.pdf` costs **2 marks** (10% of the phase), taken once from the phase total. It is not taken again from any checklist item. A mark never goes below zero.

4.9 Before you submit Phase 1

- ☐ My copy opens at `https://<username>.gblearn.com/comp1230/assignments/project/`
- ☐ I imported `schema.sql` into **my own** database, and the two sample groups show, unchanged
- ☐ I can log in as `tester1` with `Study-1230`
- ☐ My login page shows both test accounts
- ☐ An invalid group form shows an error, keeps what I typed, and saves nothing
- ☐ The Team page shows our team number, section, and every member's full name, student ID and copy link
- ☐ `aud-phase1.pdf` is in my project folder and opens
- ☐ I submitted in the D2L folder **Group Project — Phase 1** (§8)

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5 · Phase 2 · Ownership, security and joining

Due Sun 22 Nov, 11:59 p.m. (end of Week 11) · 30 marks

Phase 2 uses Week 10 (authorization and secure coding). Its full checklist is added to this brief before Week 10. What it will ask for:

- **Only the owner** can edit or delete a study group. The server checks this on every request, not only by hiding the buttons. This **replaces** the Phase 1 rule.
- A **CSRF token** on every form that changes data.
- **Join and leave.** A logged-in user can join a group they do not own, and leave a group they joined. A group page shows its members.

- **The "full" rule.** A user cannot join a group that has reached its capacity. The home page shows how many places are left.
- **My groups:** a page listing the groups the user owns and the groups they have joined.
- Your `aud-phase2.pdf`, linked from the Team page.

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6 · Phase 3 · Structure, data out, finish and deploy

Due Sun 13 Dec, 11:59 p.m. (end of Week 14) · 30 marks

Phase 3 uses Weeks 11 to 14. Its full checklist is added to this brief before Week 12. What it will ask for:

- **Structure (Week 11):** the code reorganised into the structure taught in Week 11, with the same behaviour. Some files may move outside the public folder; the Phase 3 checklist says exactly where.
- **Export data:** a group's member list downloaded as a CSV file (Week 4), and one URL that returns the upcoming study groups as JSON data (Week 12).
- **Errors (Week 13):** an error log, and a friendly error page instead of PHP error details.
- **Configuration and deployment (Week 14):** the database settings moved into a configuration file, as taught in Week 14, and the final version working on all three members' copies.
- Your `aud-phase3.pdf`, linked from the Team page.

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7 · Your video

Due Sun 13 Dec, 11:59 p.m. (end of Week 14) · 20 marks · individual

Each member makes their own short video, **3 to 5 minutes**:

1. Show **your own copy** running.
2. Open the code **you** wrote and walk through one feature, from the form to the database and back.
3. Answer **your assigned question**. Every student gets a different question about their own code, posted on **Wednesday 9 December**.

You are marked on explaining your own code correctly, not on your accent, your English or how polished the video is. Your screen and voice are required; your camera is optional.

Put the video on YouTube (unlisted is fine) or another site that plays in a browser. **Do not upload the video to your GBLearn account**; a large file can fill your space and stop your copy from working. Add the link to the Team page and submit it in the D2L folder **Group Project — Video (individual)**, with your `aud-video.pdf`.

The full video rubric is added to this brief before Week 13.

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8 · Submitting a phase

Every member submits, in the D2L folder for that phase, before the due time. The folder takes a **text submission**. Type:

```
Team N — Phase 1 completed
My copy: https://<username>.gblearn.com/comp1230/assignments/project/
My AI declaration: https://<username>.gblearn.com/comp1230/assignments/project/aud-phase1.pdf
```

- Paste addresses as plain text. D2L turns them into links.
- **Do not attach `db.php`**, and never type your database password anywhere in D2L.
- You may also attach a ZIP of your code. It is optional. **The ZIP must not contain `db.php` or any other file with a real password.** You may include a `db-example.php` with placeholder values instead.
- Your submission time is your "phase completed" time. **The checker runs on your copy in the two days after the due time** (after your submission time, if you submit late). Keep your copy as you submitted it until the instructor announces that checking for that phase is finished. After that, carry on with the next phase.

9 · AI Usage Declaration

Every member, every phase, even if you used no AI. Fill in the course's AI Usage Declaration form (Start Here → **AI Usage Declaration form**), save it as a PDF, and put it in **your own** project folder:

Phase	File	Missing or does not open
Phase 1	`aud-phase1.pdf`	−2 marks
Phase 2	`aud-phase2.pdf`	−3 marks
Phase 3	`aud-phase3.pdf`	−3 marks
Video	`aud-video.pdf`	−2 marks

Each deduction is 10% of that part's mark. It is taken once, and a mark never goes below zero. Link each file from the Team page.

AI may help you, but you must understand and be able to explain every line you submit. Your video tests this.

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10 · Team rules

- **Sharing code inside your team is expected.** Sharing code **with another team** is not allowed, and is handled under the college's academic integrity policy for both teams.
- **Agree on who does what**, and record it in your team's discussion topic.
- **Rate your team twice** in the D2L surveys **Team evaluation — check-in** (Week 11) and **Team evaluation — final** (Week 14). Only the instructor sees your answers.
- **Contribution affects marks.** If both teammates rate a member low in the final evaluation, and the video, the member's copy and the discussion topic agree, the instructor may lower that member's Phase 1–3 marks by up to 50%. The video mark is never changed this way.
- **Problems in your team?** Tell the instructor early, by Week 11 at the latest. A problem raised after a deadline is much harder to fix.